

sudo access is provided for this user with the *visudo* command. You also need to ensure that the ssh daemon is started in the guest VM using the following command:

```
$ sudo systemctl start sshd
```

You should add an entry in the */etc/hosts* file on the host system for the Parabola guest VM as indicated below:

```
192.168.122.128 parabola
```

You can now test connectivity from Ansible to the Parabola guest VM using the following Ansible commands:

```
$ ansible -i inventory/kvm/inventory parabola -m ping
parabola | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
```

Passwords

The *pwgen* utility is useful to generate passwords. It has a number of options that you can use to generate strong passwords with a combination of numerals, symbols and characters. The Ansible playbook to install *pwgen* is given below for reference:

```
- name: Harden Parabola
  hosts: parabola
  become: yes
  become_method: sudo
  gather_facts: yes
  tags: [security]

  tasks:
    - name: Install pwgen
      package:
        name: pwgen
        state: latest
```

/etc file permissions

/etc/shadow and */etc/passwd* should have restricted file permissions. The */etc/shadow* file should be owned by the *root* user and should not be accessible by any other user. The */etc/passwd* file should similarly be owned by the *root* user and have restricted permissions. The following part of the Ansible playbook checks the required permissions of these two files:

```
- name: File permissions for /etc/shadow
  stat:
    path: /etc/shadow
    register: shadow

- assert:
```

that:

```
- "shadow.stat.pw_name == 'root'"
- "shadow.stat.readable == true"
- "shadow.stat.rgrp == false"
- "shadow.stat.roth == false"
- "shadow.stat.writeable == true"
- "shadow.stat.wgrp == false"
- "shadow.stat.woth == false"
- "shadow.stat.xgrp == false"
- "shadow.stat.xoth == false"
- "shadow.stat.xusr == false"
```

```
- name: File permissions for /etc/passwd
  stat:
    path: /etc/passwd
    register: passwd
```

```
- assert:
  that:
    - "passwd.stat.pw_name == 'root'"
    - "passwd.stat.readable == true"
    - "passwd.stat.rgrp == true"
    - "passwd.stat.roth == true"
    - "passwd.stat.writeable == true"
    - "passwd.stat.wgrp == false"
    - "passwd.stat.woth == false"
    - "passwd.stat.xgrp == false"
    - "passwd.stat.xoth == false"
    - "passwd.stat.xusr == false"
```

Enforcing strong passwords

The password policy can be defined in the */etc/pam.d/passwd* file. For example, the following constraints are enforced when creating new passwords using the Ansible playbook given below:

- In case of error, prompt twice for password.
- The minimum length of the password should be ten characters.
- At least six characters should be different from the old password.
- There should be at least one digit.
- There should be at least one upper case character.
- There should be at least one other character.
- There should be at least one lower case character.

```
- name: Update /etc/pam.d/passwd
  lineinfile:
    path: /etc/pam.d/passwd
    insertbefore: '^password.*required.*pam_unix.so'
    line: 'password          required          pam_cracklib.so
retry=2 minlen=10 difok=6 dcredit=-1 ucredit=-1 ocredit=-1
lcredit=-1'
```